Task 33: Docker Compose:Real-world mutli-container App

Task 1: Build Your Own App Stack

1.Create a docker-compose.yml for a 3-service stack: A web app (use Python Flask, Node.js, or any language you know) A database (Postgres or MySQL) A cache (Redis) Write a simple Dockerfile for the web app. The app doesn't need to be complex — even a "Hello World" that connects to the database is enough.

1. create Folder: mkdir compose-app-stack , cd compose-app-stack

Create directories:

mkdir docker-compose-app
cd docker-compose-app
mkdir app
touch docker-compose.yml
cd app
touch app.py requirements.txt Dockerfile

Edit the file:`app/app.py`

from flask import Flask import psycopg2

app = Flask(**name**)

def check_db(): try: conn = psycopg2.connect(host="db", database="mydb", user="postgres", password="postgres") conn.close() return "Database Connected!" except: return "Database Not Ready"

@app.route("/") def home(): return check_db()

if **name** == "**main**": app.run(host="0.0.0.0", port=5000)

File:

```
app/requirements.txt
flask
psycopg2-binary
redis
```

Dockerfile for Web Application

app/Dockerfile

```
FROM python:3.10
WORKDIR /app
COPY requirements.txt .
RUN pip install -r requirements.txt
COPY . .
EXPOSE 5000
CMD ["python", "app.py"]
```

docker-compose.yml :

```
version: "3.9"
services:
  web:
    build: ./app
    ports:
      - "5000:5000"
    depends_on:
      db:
        condition: service_healthy
    networks:
      - mynetwork
    labels:
      project: "compose-demo"
  db:
    image: postgres:15
    environment:
      POSTGRES_DB: mydb
      POSTGRES_USER: postgres
      POSTGRES_PASSWORD: postgres
    volumes:
      - postgres_data:/var/lib/postgresql/data
    restart: always
    networks:
      - mynetwork
    healthcheck:
```

```
      test: ["CMD-SHELL", "pg_isready -U postgres"]
      interval: 5s
      retries: 5
      timeout: 3s

  redis:
    image: redis:7
    networks:
      - mynetwork
    labels:
      project: "compose-demo"

volumes:
  postgres_data:

networks:
  mynetwork:
```

Running the Application

Start the entire stack:

```
docker compose up -d
```

Check running containers:

```
docker ps
```

Expected containers:

```
web
db
redis
```

Access the application:

http://EC2_PUBLIC_IP:5000

Task 2: depends_on & Healthchecks Add depends_on to your compose file so the app starts after the database Add a healthcheck on the database service Use depends_on with condition:

service_healthy so the app waits for the database to be truly ready, not just started Test: Bring everything down and up — does the app wait for the DB?

Service Dependency with Health Checks

The web service waits for the database to become healthy.

```
depends_on:
  db:
    condition: service_healthy
```

Database healthcheck:

```
healthcheck:
  test: ["CMD-SHELL", "pg_isready -U postgres"]
  interval: 5s
  retries: 5
```

This ensures the web service starts **only after PostgreSQL is ready**.
Task 3: Restart Policies Add restart: always to your database service Manually kill the database container — does it come back? Try restart: on-failure — how is it different? Write in your notes: When would you use each restart policy?

Restart Policies : Database service includes:

```
restart: always
```

Testing Restart Policy

Find container ID:

```
docker ps
```

Kill database container:

```
docker kill <container-id>
```

Check again:

```
docker ps
```

The container automatically restarts.

Task 4: Custom Dockerfiles in Compose Instead of using a pre-built image for your app, use build: in your compose file to build from a Dockerfile Make a code change in your app Rebuild and restart with one command
=> Rebuilding After code chnages
If application code changes, rebuild the image.
Docker compose up –build -d

Task 5: Named Networks & Volumes Define explicit networks in your compose file instead of relying on the default Define named volumes for database data Add labels to your services for better organization

=> Named volume used for database persistence:`postgres_data`

Check volumes:`docker volume ls`

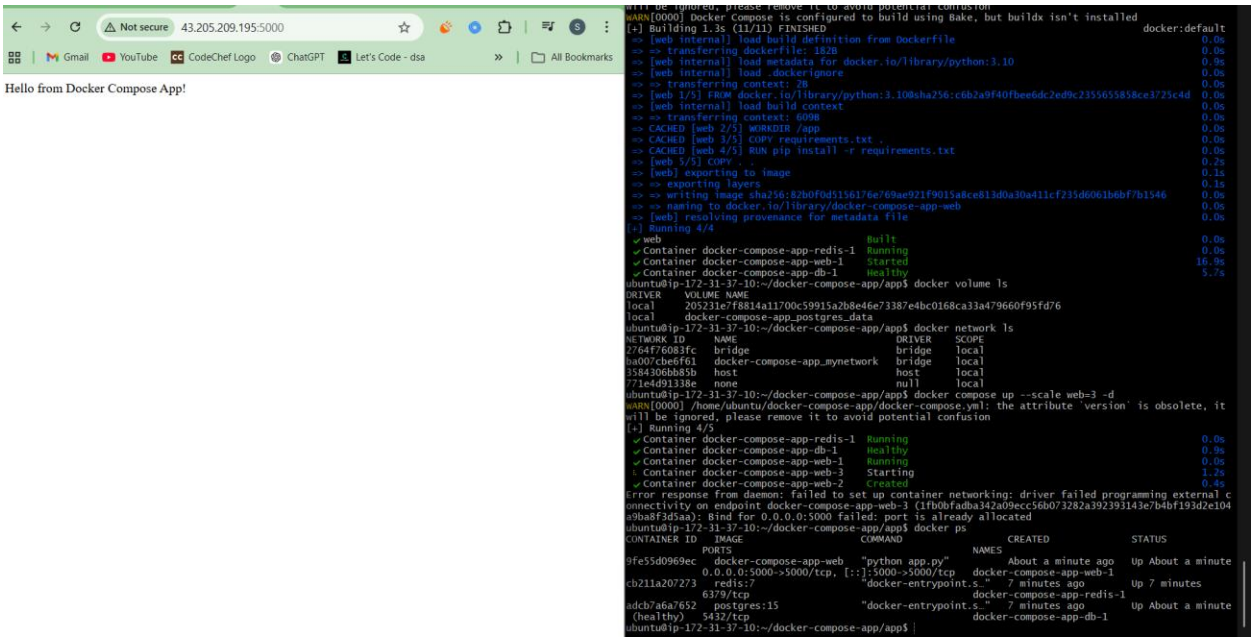
Custom network:`mynetwork`

Check networks:`docker network l`

Task 6: Scaling (Bonus) Try scaling your web app to 3 replicas using docker compose up --scale What happens? What breaks? Write in your notes: Why doesn't simple scaling work with port mapping?

=> Scale web application:`docker compose up --scale web=3 -d`

Check containers:`docker ps`


Not secure 43.205.209.195:5000
Gmail YouTube CodeChef Logo ChatGPT Let's Code - dsa All Bookmarks
Hello from Docker Compose App!
will be ignored, please remove it to avoid potential confusion
WARN[0000] Docker Compose is configured to build using Bake, but buildx isn't installed
[+] Building 1.3s (11/11) FINISHED docker:default
=> [web internal] load build definition from Dockerfile 0.0s
=> => transferring dockerfile: 182B 0.0s
=> [web internal] load metadata for docker.io/library/python:3.10 0.9s
=> [web internal] load .dockerignore 0.0s
=> => transferring context: 2B 0.0s
=> [web 1/5] FROM docker.io/library/python:3.10@sha256:c6b2a9f40fbee6dc2ed9c2355655858ce3725c4d 0.0s
=> [web internal] load build context 0.0s
=> => transferring context: 609B 0.0s
=> CACHED [web 2/5] WORKDIR /app 0.0s
=> CACHED [web 3/5] COPY requirements.txt . 0.0s
=> CACHED [web 4/5] RUN pip install -r requirements.txt 0.0s
=> [web 5/5] COPY . . 0.2s
=> [web] exporting to image 0.1s
=> => exporting layers 0.1s
=> => writing image sha256:82b0f0d5156176e769ae921f9015a8ce813d0a30a411cf235d6061b6bf7b1546 0.0s
=> => naming to docker.io/library/docker-compose-app-web 0.0s
=> [web] resolving provenance for metadata file 0.0s
[+] Running 4/4
✔ web Built 0.0s
✔ Container docker-compose-app-redis-1 Running 0.0s
✔ Container docker-compose-app-web-1 Started 16.9s
✔ Container docker-compose-app-db-1 Healthy 5.7s
ubuntu@ip-172-31-37-10:~/docker-compose-app/app$ docker volume ls
DRIVER VOLUME NAME
local 205231e7f8814a11700c59915a2b8e46e73387e4bc0168ca33a479660f95fd76
local docker-compose-app_postgres_data
ubuntu@ip-172-31-37-10:~/docker-compose-app/app$ docker network ls
NETWORK ID NAME DRIVER SCOPE
2764f76083fc bridge bridge local
ba007cbe6f61 docker-compose-app_mynetwork bridge local
3584306bb85b host host local
771e4d91338e none null local
ubuntu@ip-172-31-37-10:~/docker-compose-app/app$ docker compose up --scale web=3 -d
WARN[0000] /home/ubuntu/docker-compose-app/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
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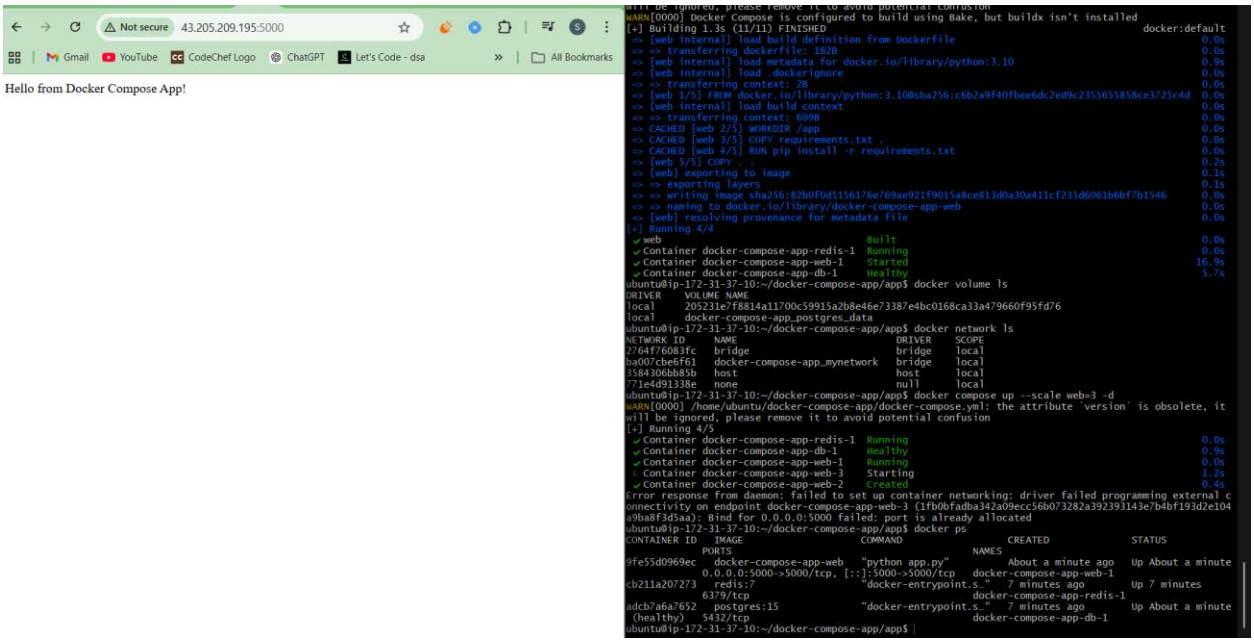
✔ Container docker-compose-app-redis-1 Running 0.0s
✔ Container docker-compose-app-db-1 Healthy 0.9s
✔ Container docker-compose-app-web-1 Running 0.0s
⠼ Container docker-compose-app-web-3 Starting 1.2s
✔ Container docker-compose-app-web-2 Created 0.4s
Error response from daemon: failed to set up container networking: driver failed programming external connectivity on endpoint docker-compose-app-web-3 (1fb0bfadba342a09ecc56b073282a392393143e7b4bf193d2e104a9ba8f3d5aa): Bind for 0.0.0.0:5000 failed: port is already allocated
ubuntu@ip-172-31-37-10:~/docker-compose-app/app$ docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
9fe55d0969ec docker-compose-app-web "python app.py" About a minute ago Up About a minute 0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp docker-compose-app-web-1
cb211a207273 redis:7 "docker-entrypoint.s…" 7 minutes ago Up 7 minutes 6379/tcp docker-compose-app-redis-1
adcb7a6a7652 postgres:15 "docker-entrypoint.s…" 7 minutes ago Up About a minute (healthy) 5432/tcp docker-compose-app-db-1
ubuntu@ip-172-31-37-10:~/docker-compose-app/app$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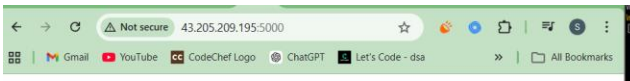

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local     docker-compose-app_postgres_data
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NETWORK ID     NAME                          DRIVER    SCOPE
2764f76083fc   bridge                        bridge    local
ba007cbe6f61   docker-compose-app_mynetwork  bridge    local
3584306bb85b   host                          host      local
771e4d91338e   none                          null      local
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CONTAINER ID   IMAGE                     COMMAND                  CREATED              STATUS
              PORTS                                           NAMES
9fe55d0969ec   docker-compose-app-web   "python app.py"          About a minute ago   Up About a minute
              0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp   docker-compose-app-web-1
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              6379/tcp                                        docker-compose-app-redis-1
adcb7a6a7652   postgres:15              "docker-entrypoint.s…"   7 minutes ago        Up About a minute
 (healthy)   5432/tcp                                         docker-compose-app-db-1
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```